

Computació Gràfica i Multimèdia

Work package 1

Master's Degree in Informatics Engineering

2026/2027 year

This document describes the tasks to carry out as part of work package 1.

Example

Together with this document, you are provided a script file "Maze.cs". You can test it as follows:

1. Create a 2D project in Unity.
2. Create a 2D square and color it white. Name it "Corridor" and create an asset from it.
3. Create a 2D square and color it blue. Name it "Wall" and create an asset from it.
4. Import the "Maze.cs" script into the assets area.
5. Create an empty object, name it "GameControl" and make the "Maze.cs" script be a component of it.
6. Drag the "Corridor" and "Wall" assets into the slots available inside the "Maze.cs" script in the inspector.
7. Run the game. After adjusting the camera size, you should see a result similar to:



The "Maze.cs" script takes a 2D array representing a tiled world. Each element of the array represents a tile and its content determines whether it is of "Wall" or "Corridor" type. The provided example assigns the border tiles to be of "Wall" type while the remaining ones but a randomly chosen one, are set to be of "Corridor" type.

Task 1. Generate a random maze

You are asked to extend the “Maze.cs” script so that it generates and displays a random maze. If you prefer, you can discard the example and start a new project from scratch.



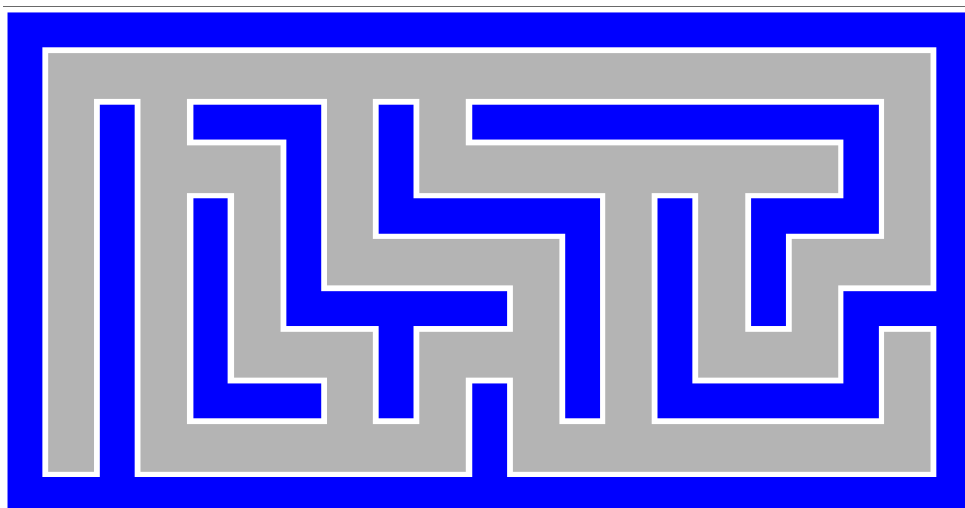
The requirements are the following:

- The resulting maze must be connected. Namely, there exists a path between any pair of corridor-type cells.
- Implement it so that it is easy to modify the dimensions (number of rows and columns) of the generated maze.
- The resulting maze must include some loops.

Task 2. Use of ruled tile maps

Make use of ruled tile maps in your project. The walls of the randomly generated maze must be drawn using a ruled tile map.

- You can create your own sprite sheet or import it from the Internet.



Evaluation

The mark for this task is computed as:

- 80% : tasks detailed in this document
- 20% : additional contributions of your choice

The additional contributions of your choice should not take more than 20% of the time you devoted to this work package.

Submission

This task is presented as follows:

- Prepare a PDF document containing some snapshots showing that you have properly completed the task and upload it to the activity created in the virtual campus.
- Show your completed work package to the professor who will ask you some questions.